Hybrid WAN (SD-WAN) : Introduction to SD-WAN, Key Benefits of SD-WAN

1. Create a comparison table listing at least 5 key differences between traditional WAN and SD-WAN, focusing on how data is routed, managed, and secured..

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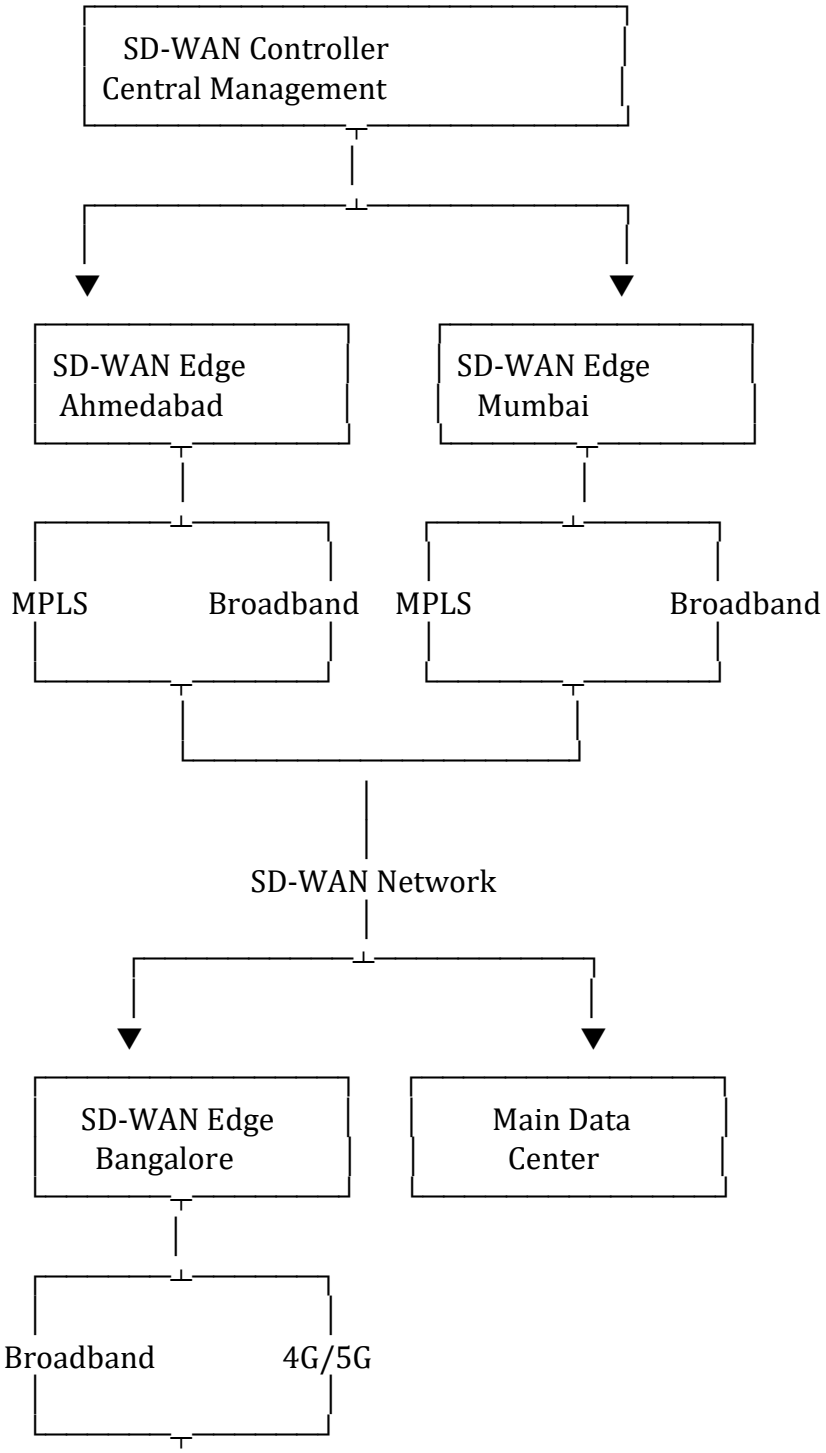
- Comparison Between Traditional WAN and SD-WAN

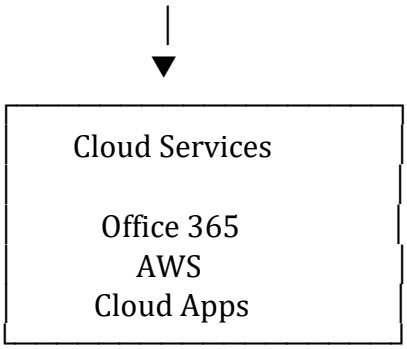
No.	Traditional WAN	SD-WAN
1. Data Routing	Traffic is usually routed through fixed paths, often using MPLS circuits.	Traffic can be dynamically routed through MPLS, broadband, 4G/5G, or other available links based on network conditions.
2. Network Management	Network devices are generally configured and managed individually at each branch.	Network policies and configurations can be managed centrally using an SD-WAN controller.
3. Path Selection	The network generally uses predefined routes and has limited ability to react to changing network conditions.	SD-WAN continuously evaluates link conditions such as latency, packet loss, and bandwidth and can select a better path.
4. Security	Security may depend heavily on separate firewalls and security devices at branch locations.	SD-WAN can integrate security features such as encryption, segmentation, firewalls, and secure tunnels.
5. Cloud Access	Branch traffic may need to travel through the main data center before reaching cloud applications.	SD-WAN can provide direct and optimized access from branches to cloud applications.
6. Cost	Dedicated MPLS connections can be expensive, especially for multiple branches.	Organizations can combine lower-cost broadband and other links with MPLS to reduce connectivity costs.
7. Scalability	Adding new branches can require significant manual configuration and new dedicated circuits.	New branches can be deployed more quickly using centralized configuration and multiple types of Internet connections.

2. Draw a simple network diagram (on paper or using a tool like draw.io) showing how SD-WAN connects multiple branch offices to cloud apps (like Office 365 or AWS) and the main data center. Label the SD-WAN controllers, edges, and at least two types of WAN links (e.g., MPLS, broadband).

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- SD-WAN Network Diagram





- **Explanation:**

1. **SD-WAN Controller:**
   The SD-WAN controller provides centralized management of the SD-WAN network. Administrators can configure policies, routing rules, security policies, and monitor network performance from a central location.
2. **SD-WAN Edge:**
   An SD-WAN edge device is installed at each branch office. It connects the branch network to different WAN links and forwards traffic according to SD-WAN policies.
3. **MPLS:**
   MPLS can provide a reliable connection for important business applications and traffic.
4. **Broadband:**
   Broadband Internet provides an additional and generally lower-cost WAN connection. SD-WAN can use it when appropriate or as a backup to MPLS.
5. **4G/5G:**
   A mobile network connection can provide another backup connection if the primary WAN connections become unavailable.
6. **Cloud Applications:**
   Applications such as Microsoft 365 and AWS can be accessed directly through optimized SD-WAN paths instead of always sending traffic through the main data center.
7. **Main Data Center:**
   The data center contains the organization's internal servers, applications, databases, and other important resources.

3. Pick any app you use daily (like WhatsApp, Zomato, or Instagram) and write a short scenario explaining how SD-WAN could improve user experience for that app when used across multiple cities in India. Hint: Think about speed, reliability, and security when users access the app from different locations.

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- SD-WAN Scenario for Zomato

  Suppose Zomato operates offices and technical infrastructure in cities such as Ahmedabad, Mumbai, Bangalore, Delhi, and Hyderabad. Employees in these locations need to access Zomato's cloud applications, databases, communication systems, and other business services.

  Without SD-WAN, a branch office may have to use a fixed WAN path to reach applications. If that connection becomes slow or experiences packet loss, employees may experience slow application performance.

  With SD-WAN, the network can continuously monitor different connections such as MPLS, broadband, and 4G/5G. For example, if the broadband connection between the Ahmedabad office and a cloud application becomes slow, SD-WAN can route important traffic through another available connection.

  For employees using Zomato's internal applications, this can provide better performance and reliability. SD-WAN can also use encryption and security policies to protect traffic between offices and cloud services.

  Therefore, SD-WAN can improve the user experience by providing better speed, reliable connectivity, automatic path selection, and secure communication across different cities in India.

4. List and explain three key benefits of SD-WAN for a company with offices in Ahmedabad, Mumbai, and Bangalore. For each benefit, give a real-world example of how it would help employees or customers.

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- **Three Key Benefits of SD-WAN**

1. Better Reliability and Automatic Failover
   SD-WAN can use multiple WAN connections such as MPLS, broadband, and 4G/5G. If one connection fails or becomes unstable, SD-WAN can automatically move traffic to another available connection.

   Real-world example:
   Suppose employees in the Ahmedabad office are using the company's cloud application and the primary MPLS connection fails. SD-WAN can automatically route the traffic through the broadband connection so employees can continue working with minimal interruption.

2. Improved Application Performance
   SD-WAN can identify applications and select an appropriate network path based on factors such as latency, packet loss, and available bandwidth. This is especially useful for cloud applications and real-time communication.

   Real-world example:
   Employees in the Mumbai office are attending an important video conference with the Bangalore office. If one WAN link has high latency or packet loss, SD-WAN can route the video conferencing traffic through a better-performing link, helping reduce interruptions and poor call quality.

3. Centralized Management and Improved Security

   SD-WAN allows network administrators to manage network policies, routing, and security from a central management platform. This makes it easier to manage multiple offices and maintain consistent policies.

   Real-world example:

   Suppose a company has offices in Ahmedabad, Mumbai, and Bangalore. The network administrator can create a security and access policy centrally and apply it to all three offices. If the company adds another branch, the administrator can apply the same policies to the new location without manually configuring every network device.